

```
In [1]: %load_ext autoreload
%autoreload 2

# Control figure size
figsize=(15, 4)

from sklearn.neighbors import KernelDensity
from sklearn.model_selection import GridSearchCV
from util import util
import numpy as np
from matplotlib import pyplot as plt
import pandas as pd
import warnings
warnings.filterwarnings(action='once')
warnings.filterwarnings('ignore')
```

Gaussian Processes

If we want to do better, we need a better predictor
What would be some desired properties?

Filling Values Using a Model

Given a gap (i.e. one or more contiguous missing values), the model:

- Should be able to make a prediction about the missing values
- It should take into account the values that are *before* the hole
- ...But also those that are *after* the hole

I.e. it should be able to interpolate *all the available data*

Can you think of an ML model that can to that?

If you have trouble, that's normal: there aren't many!

- Most classical ML models *cannot* be used for filling (at least not easily)
- We will see a couple of examples now

Filling Values Using a Density Estimator

Density estimation does take into account all data

...But it cannot easily provide predictions

To be fair, predictions *can be obtained* from a density estimator

- Given an estimator $f(x, \theta)$ for $P(x)$
- ...Then we can find the most likely value for x by solving:

$$\operatorname{argmax}_x f(x, \theta)$$

- This is called a *Maximum A Posteriori (MAP)*

In fact, it's what we naturally get out of most regressors/classifiers

...But with density estimators, computing the MAP can be *very expensive*

Filling Values Using a Regressor

We can train a (auto)-regressor to predict the next value in a series

...Then we can fill holes by making a "*rolling forecast*"

- We make one prediction
- We incorporate the prediction in the input
- Then we make another prediction and so on

...But by doing so we only rely on *past* values

In fact, the last prediction can be far from the next true value

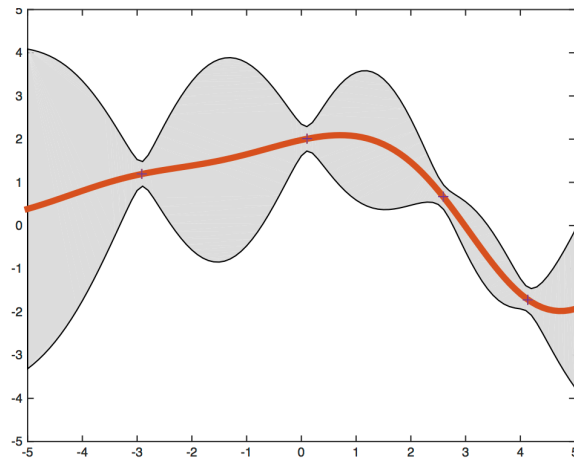
- In addition, every iteration of the process introduces approximation
- Formally, we get *compound errors*

This is partly unavoidable: predicting the far future is often difficult

Gaussian Processes

One of the few viable ML models is given by *Gaussian Processes (GP)*

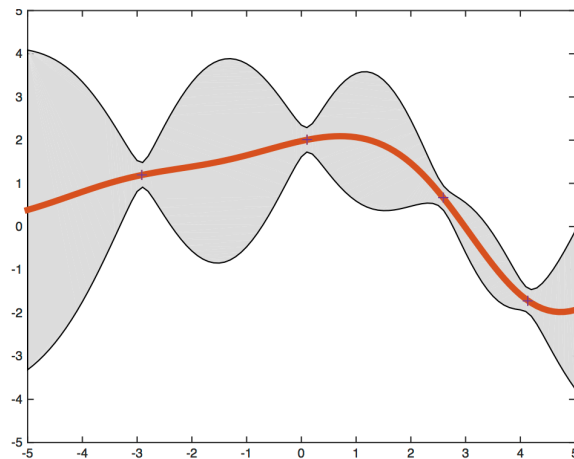
They are not easy to use, but can work well in the correct setting



- GPs define a distribution mainly conditioned by an *index (input) variable*
- The distribution is based on *the available observations* and *a few assumptions*

Gaussian Processes

One of the few viable ML models is given by *Gaussian Processes (GP)*

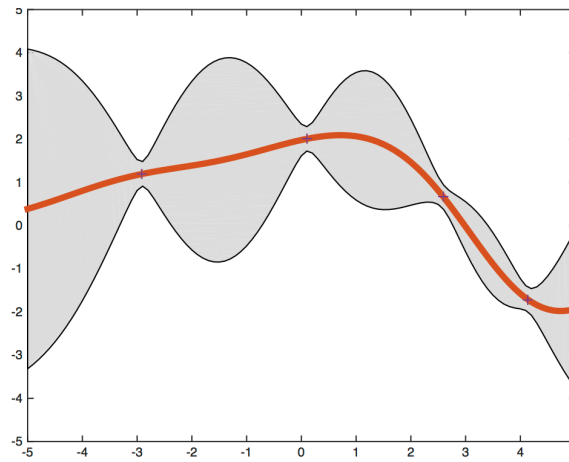


Assumption 1 (intuitively)

- *For every value* of the input variable the distribution is *Gaussian*
- Therefore it can be described by a *mean* and a *standard deviation*

Gaussian Processes

One of the few viable ML models is given by *Gaussian Processes (GP)*



Assumption 2 (intuitively)

- The stdev depends on the distance between a point and the observations
- So it will be *low* when we are *close* to the observations, *high* when we are *far*

Gaussian Processes

Formally, things are a bit more complicated

A GP is a *stochastic process*, i.e. a collection of indexed random variables

- The index variable x represents an input (and can be a vector/tuple)
- Each variable y_x represents the output for input x
- The index is *continuous* and the collection is therefore *infinite*

You can think of y_x as the value of a (stochastic) function for input x

Some examples:

- y_x could be the rainfall rate at location $x = (\text{latitude}, \text{longitude})$
- y_x could be the traffic volume at time x

Each y_x follows a Normal Distribution, but the variables are *correlated*

- Therefore every finite subset of y_x variables
- ...Follows a *Multivariate Normal Distribution*

Multivariate Normal Distribution

Why the multivariate normal distribution?

- It works for many real world phenomena
- It has a (relatively) simple [closed-form density function](#)

In particular, the PDF for a MND is defined via:

- A (vector) mean μ
- A covariance matrix Σ

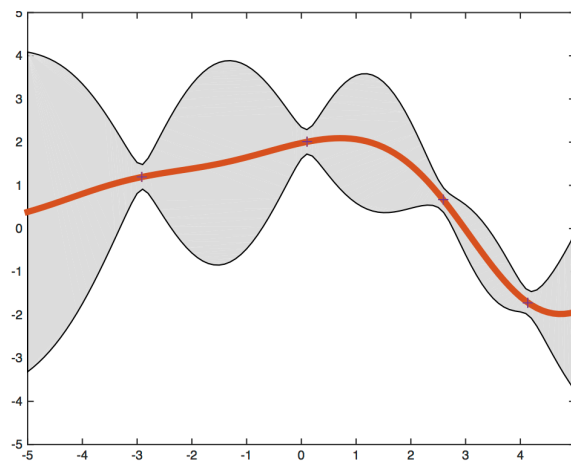
By recentering we can assume $\mu = 0$, meaning that *knowing Σ is enough*

Therefore, if we know Σ we can easily compute

- The joint density $f(\bar{y}_x)$ for a set of observations
- The conditional density $f(y_x | \bar{y}_x)$ of a new observation y_x , given $\bar{y}_x$

Why do We Care?

We need the conditional density to perform inference!



- The line and grey areas represent the conditional density $f(y_x | \bar{y}_x)$ of y_x
- ...Based on the available observations, i.e. $\bar{y}_x$

Why do We Care

We need the joint density to perform training!

...Because in practice we *don't know* Σ

- Therefore we'll assume that Σ is a *parameterized function* $\Sigma(\theta)$
- ...And we can optimize the parameters θ for maximum likelihood

Formally, given a set of training observations $\bar{y}_x$

...We can calibrate the parameters by solving a problem in the form:

$$\operatorname{argmax}_{\theta} f(\bar{y}_x, \theta)$$

- Here we are not using a product of probabilities over the training set

- ...Because the y_x variables are correlated
- No worries: since we have the joint PDF, we use it directly

Which Covariance Matrix?

We still have one big problem

Say that we have a covariance matrix Σ for a set of observations $\bar{y}_x$

- Now we want to perform inference for an input value x
- Formally: we want to compute $f(y_x | \bar{y}_x)$

In principle, we can use the formula:

$$f(y_x | \bar{y}_x) = \frac{f(y_x, \bar{y}_x)}{f(\bar{y}_x)}$$

- By using our Σ , we can easily compute $f(\bar{y}_x)$
- ...But what about $f(y_x, \bar{y}_x)$?

Which Covariance Matrix?

Our Σ refers to our set of observed variables $\bar{y}_x$

Therefore, if we have n variables, our matrix will be $n \times n$

$$\Sigma_x = \begin{pmatrix} \sigma_{\bar{x}_1, \bar{x}_1} & \sigma_{\bar{x}_1, \bar{x}_2} & \cdots & \sigma_{\bar{x}_1, \bar{x}_n} \\ \sigma_{\bar{x}_2, \bar{x}_1} & \sigma_{\bar{x}_2, \bar{x}_2} & \cdots & \sigma_{\bar{x}_2, \bar{x}_n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{\bar{x}_n, \bar{x}_1} & \sigma_{\bar{x}_n, \bar{x}_2} & \cdots & \sigma_{\bar{x}_n, \bar{x}_n} \end{pmatrix}$$

- In every cell we have the covariance for variables $\bar{y}_{x_i}$ and $\bar{y}_{x_j}$
- With this matrix, we can compute $f(\bar{y}_x)$

Which Covariance Matrix?

However, $f(y_x, \bar{y}_x)$ refers to *one more variable*

Meaning that it will be specified via an $(n + 1) \times (n + 1)$ matrix!

$$\Sigma_{x,\bar{x}} = \begin{pmatrix} \sigma_{x,x} & \sigma_{x,\bar{x}_1} & \sigma_{x,\bar{x}_2} & \cdots & \sigma_{x,\bar{x}_n} \\ \sigma_{\bar{x}_1,x} & \sigma_{\bar{x}_1,\bar{x}_1} & \sigma_{\bar{x}_1,\bar{x}_2} & \cdots & \sigma_{\bar{x}_1,\bar{x}_n} \\ \sigma_{\bar{x}_2,x} & \sigma_{\bar{x}_2,\bar{x}_1} & \sigma_{\bar{x}_2,\bar{x}_2} & \cdots & \sigma_{\bar{x}_2,\bar{x}_n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_{\bar{x}_n,x} & \sigma_{\bar{x}_n,\bar{x}_1} & \sigma_{\bar{x}_n,\bar{x}_2} & \cdots & \sigma_{\bar{x}_n,\bar{x}_n} \end{pmatrix}$$

- Assuming that $\bar{y}_{\bar{x}}$ are the training observations
- ...We could define $\sigma_{\bar{x}_1,\bar{x}_2}$ at training time

But how do we define *the new covariances*, i.e. those related to y_x ?

Defining the Covariance Matrix

We assume that covariance can be *built from a set of inputs*

Let x refer now to a vector of values for the index variable

- Given two variables y_{x_i} and y_{x_j}
- We specify their covariance via parameterized *kernel function* $K_\theta(x_i, x_j)$
- K typically depends on the distance between input values

Given any finite set of variables $\{y_{x_1}, \dots, y_{x_n}\}$, the covariance matrix is:

$$\Sigma = \begin{pmatrix} K_\theta(x_1, x_1) & K_\theta(x_1, x_2) & \cdots & K_\theta(x_1, x_n) \\ K_\theta(x_2, x_1) & K_\theta(x_2, x_2) & \cdots & K_\theta(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ K_\theta(x_n, x_1) & K_\theta(x_n, x_2) & \cdots & K_\theta(x_n, x_n) \end{pmatrix}$$

...Which we can compute based on the input (and the parameters) alone!

The Whole Picture

In practice, at *training time*

- Pick a *parameterized* kernel function $K_\theta(x_i, x_j)$
- Collect training observations $\bar{y}_{\bar{x}}$
- Optimize the kernel for maximum likelihood (e.g. via gradient descent)

Both the parameters θ and the observations $\{\bar{y}_{\bar{x}}\}$ are stored in the model

- This is similar to what we have in Kernel Density Estimation

At inference time:

- Given a new input (i.e. index) value x
- We obtain the covariance matrix $\Sigma_{\bar{x}}$
- We obtain the covariance matrix $\Sigma_{x,\bar{x}}$

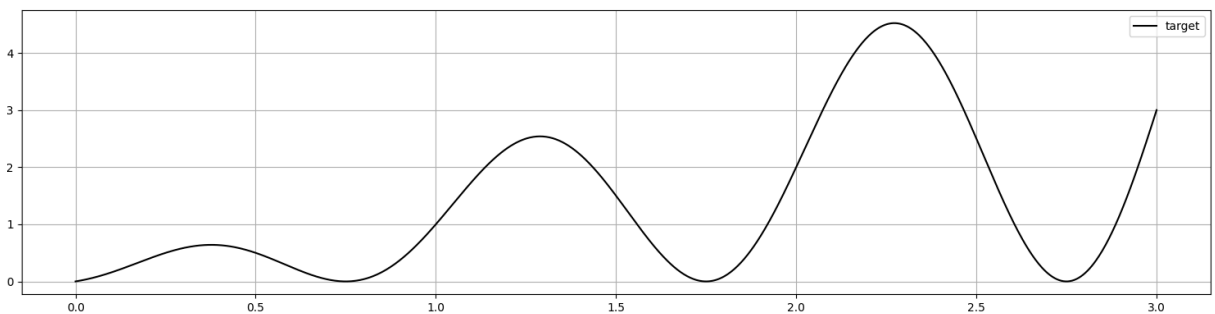
...And with this we can completely characterize $f(y_x, | \bar{y}_{\bar{x}})$

Gaussian Processes in scikit-learn

Let's see how to use Gaussian Processes in scikit-learn

First, let us choose a function as our ground truth

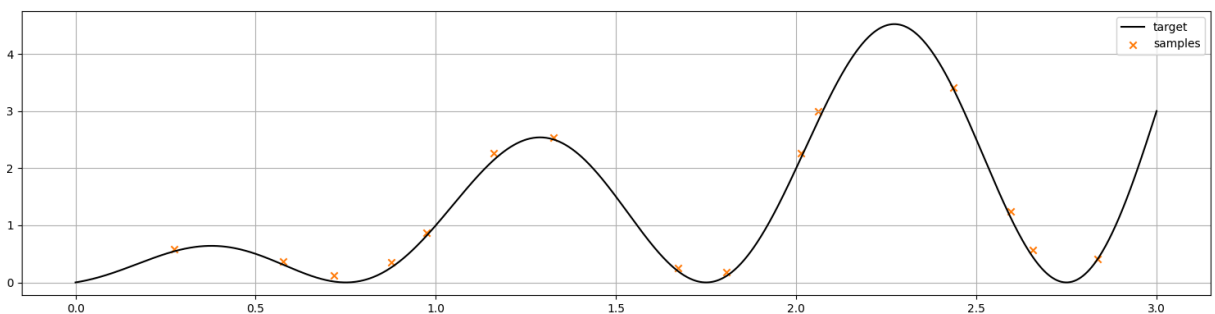
```
In [2]: f = lambda x: x * np.sin(2*np.pi*x) + x # target function
x = np.linspace(0, 3, 1000)
y = pd.Series(index=x, data=f(x))
util.plot_gp(target=y, figsize=figsize)
```



Gaussian Processes in scikit-learn

Then we build a small training set:

```
In [3]: np.random.seed(42)
n_tr = 15
x_tr = np.linspace(0.2, 2.8, n_tr) + 0.2*np.random.rand(n_tr)
x_tr.sort()
y_tr = pd.Series(index=x_tr, data=f(x_tr) + 0.2*np.random.rand(n_tr))
util.plot_gp(target=y, samples=y_tr, figsize=figsize)
```



Gaussian Processes in scikit-learn

Now we are where we would normally start in a data analysis

...And we need to choose a kernel among are [many available options](#)

- Since we have 15 training points
- This will be used at training time to build a 15×15 covariance matrix

We will start with a simple **Radial Basis Function (i.e. Gaussian) kernel**

$$K(x_i, x_j) = e^{-\frac{d(x_i, x_j)^2}{2l}}$$

The covariance *decreases with the (Euclidean) distance* $d(x_i, x_j)$:

- Intuitively, *the closer the points, the higher the correlation*
- The l parameter (*scale*) control the rate of the reduction

Gaussian Processes in scikit-learn

Here's how to use an RBF kernel in scikit-learn

```
In [4]: from sklearn.gaussian_process.kernels import RBF
kernel = RBF(1, (1e-2, 1e2))
```

The RBF kernel has a single parameter, representing its *scale*

The extra (tuple) parameter represents a pair of *bounds*

- During training, only values within the boundaries will be considere

Bounds can be very useful for controlling the training process

- They should be chosen based on the available domain information
- Multiples educated guesses are fine when little is known about the problem

Gaussian Processes in scikit-learn

Now we can train a Gaussian Process

```
In [5]: from sklearn.gaussian_process import GaussianProcessRegressor
gp = GaussianProcessRegressor(kernel=kernel, n_restarts_optimizer=9)
gp.fit(y_tr.index.values.reshape(-1,1), y_tr.values) # needs 2D input
```

```
Out[5]: ▼ GaussianProcessRegressor ⓘ ?
        ► Parameters
```

- Training uses Gradient Descent to maximize the likelihood of the training data
- *Restarts* are needed to mitigate issues due to local optima

Training can be an expensive process:

- Building the covariance matrix runs in $O(n^2)$ (where n is the training set size)
- ...And we need to do that repeatedly during optimization!

Gaussian Processes in scikit-learn

Finally, we can obtain the predictions:

```
In [6]: xp, std = gp.predict(x.reshape(-1,1), return_std=True)
xp = pd.Series(index=y.index, data=xp)
std = pd.Series(index=y.index, data=std)
```

The "predictions" are *not point estimates*!

- We actually get those (i.e. the means, called `xp` in the code)
- ...But we also obtain standard deviations

The model output is a fully characterized conditional distribution

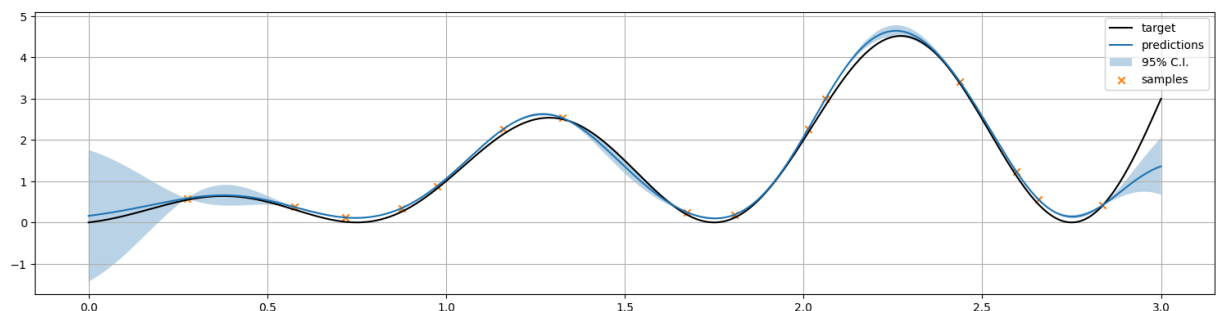
The distribution is conditional on:

- The input value x (passed at inference time)
- All the training observations (stored in the model)

Gaussian Processes in scikit-learn

We can now plot the predictions

```
In [7]: util.plot_gp(target=y, samples=y_tr, pred=xp, std=std, figsize=figsize)
```



Obtaining a full distribution as output is cool

- We obtain an estimated value (the mean)

- ...And we get to know how certain the model is about that

Behind the Scenes

Let's see how prediction works with a numeric example

We want a prediction for $x = 2.5$, i.e. $y_{2.5}$

- We consider two separate training set (and pre-optimized kernel parameters)
- The first set contains only $(\bar{x}_9, \bar{y}_{\bar{x}_9}) \simeq (2.01, 2.27)$
- The second set contains only $(\bar{x}_0, \bar{y}_{\bar{x}_0}) \simeq (0.27, 0.58)$

The covariance matrices in the two cases are therefore:

$$\Sigma_{y_x, \bar{y}_{\bar{x}_9}} = \begin{pmatrix} K(2.01, 2.01) & K(2.01, 2.5) \\ K(2.5, 2.01) & K(2.5, 2.5) \end{pmatrix}$$

$$\Sigma_{y_x, \bar{y}_{\bar{x}_0}} = \begin{pmatrix} K(0.27, 0.27) & K(0.27, 2.5) \\ K(2.5, 0.27) & K(2.5, 2.5) \end{pmatrix}$$

Behind the Scenes

Let's actually build the matrices in Python

- Note: scikit-learn kernels are not designed to be used on individual points
- So, for this we will rely on basic numpy methods

We start with $\bar{x}_9$ and x , which are *close to each other*

```
In [8]: from scipy.stats import multivariate_normal
X9, X0, X = [[x_tr[9]], [x_tr[0]], [2.5]] # Must be 2D
sigma_9x = np.array([[kernel(X9, X9)[0,0], kernel(X9, X)[0,0]],
                    [kernel(X, X9)[0,0], kernel(X, X)[0,0]])
f_9x = multivariate_normal([0, 0], cov=sigma_9x)
```

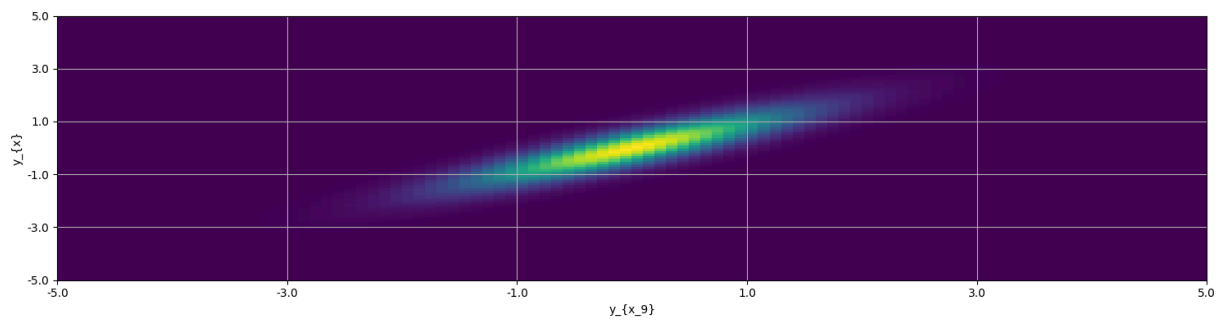
Then we do the same for $\bar{x}_0$ and x , which are *far apart*

```
In [9]: sigma_0x = np.array([[kernel(X0, X0)[0,0], kernel(X0, X)[0,0]],
                             [kernel(X, X0)[0,0], kernel(X, X)[0,0]])
f_0x = multivariate_normal([0, 0], cov=sigma_0x)
```

Behind the Scenes

$\bar{x}_9$ and x are *close to each other*, so $\bar{y}_{\bar{x}_9}$ and y_x are *strongly correlated*

```
In [10]: yr = np.linspace(-5, 5, 100)
util.plot_distribution_2D(f_9x, yr, yr, figsize=figsize)
plt.xlabel('y_{x_9}'); plt.ylabel('y_{x}'); plt.tight_layout()
```

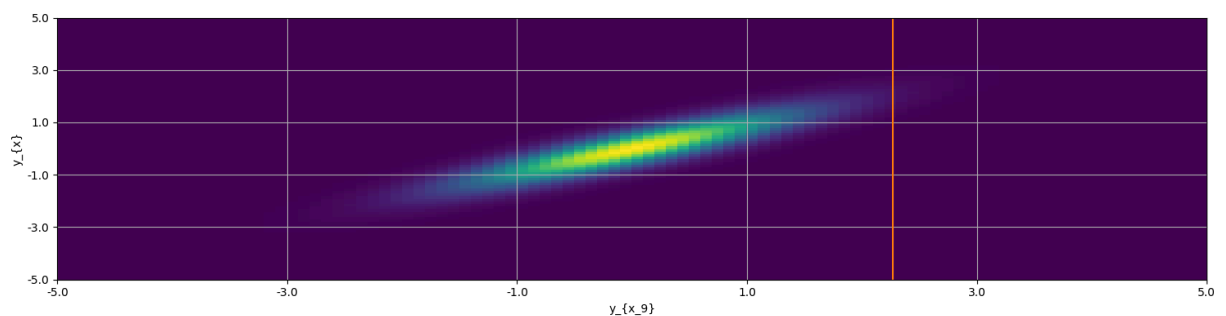


- If we know *neither* $\bar{y}_{x_0}$ nor y_x , we can only say that they are likely *both* zero

Behind the Scenes

But we *do know* $\bar{y}_{\bar{y}_9}$! So, we can use this information

```
In [11]: util.plot_distribution_2D(f_9x, yr, yr, figsize=figsize)
plt.axvline(10*(y_tr[x_tr[9]] + 5), color='tab:orange');
plt.xlabel('y_{x_9}'); plt.ylabel('y_{x}'); plt.tight_layout()
```

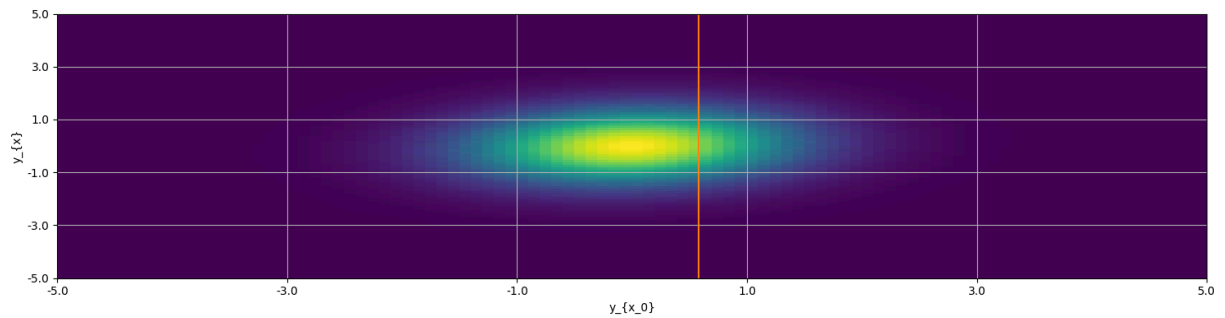


- Given the observation, the most likely value for y_x is $\simeq 2$

Behind the Scenes

$\bar{x}_0$ and x are *far apart*, so $\bar{y}_{x_0}$ and y_x are *loosely correlated*

```
In [12]: util.plot_distribution_2D(f_0x, yr, yr, figsize=figsize)
plt.axvline(10*(y_tr[x_tr[0]] + 5), color='tab:orange');
plt.xlabel('y_{x_0}'); plt.ylabel('y_{x}'); plt.tight_layout()
```



- Knowing $\bar{y}_{x_0}$ is not going to be of much help here

Memento

So, a few key insight to keep in mind:

- Superficially, GPs behave like *functions that output probability distribution*
- Internally, they store *the kernel*, defining how all the points are correlated
- *A set of observations*, used to obtain conditional distributions

In scikit-learn:

When we call the `fit` method:

- The optimizer *adjusts the kernel parameters*
- ...And *the observations* $\bar{y}_{\bar{x}}$ are stored_

When we call the `predict` method:

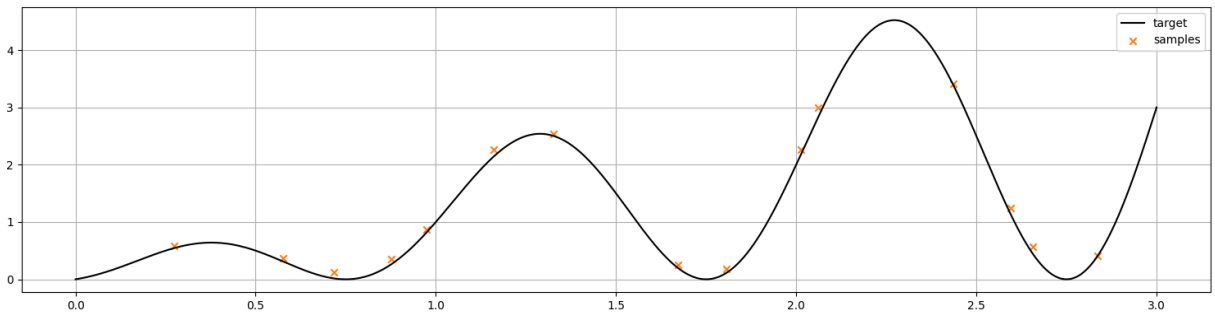
- The covariance matrix is built
- The model computes the conditional distributions

How to Improve the Model

We can improve the model by choosing a more appropriate kernel

- Let's (cheat and) look again at our ground truth and training data
- Which patterns and properties do you notice?

```
In [13]: util.plot_gp(target=y, samples=y_tr, figsize=figsize)
```



We have some *noise*, a *period*, and a *trend*

How to Improve the Model

So, let us deal with the noise first

```
In [14]: from sklearn.gaussian_process.kernels import WhiteKernel

kernel = WhiteKernel(0.1, (1e-2, 1e2))
kernel += RBF(1, (1e-2, 1e2))
```

WhiteKernel captures the presence of *noise* in the data

$$K(x_i, x_j) = \sigma^2 \text{ iff } x_i = x_j, 0 \text{ otherwise}$$

- The only parameter of **WhiteKernel** represents the noise level σ^2
- A small noise level prevents overfitting
- ...But too much noise leads to useless predictions!

How to Improve the Model

It's often a good idea to have *magnitude parameters* in the kernel

```
In [15]: from sklearn.gaussian_process.kernels import ConstantKernel

kernel = WhiteKernel(0.1, (1e-2, 1e2))
kernel += ConstantKernel(1, (1e-2, 1e2)) * RBF(1, (1e-2, 1e2))
```

ConstantKernel is a constant factor (in this case a relative weight)

- ...And allows the optimizer to tune the magnitude of the RBF kernel

Let's repeat training again:

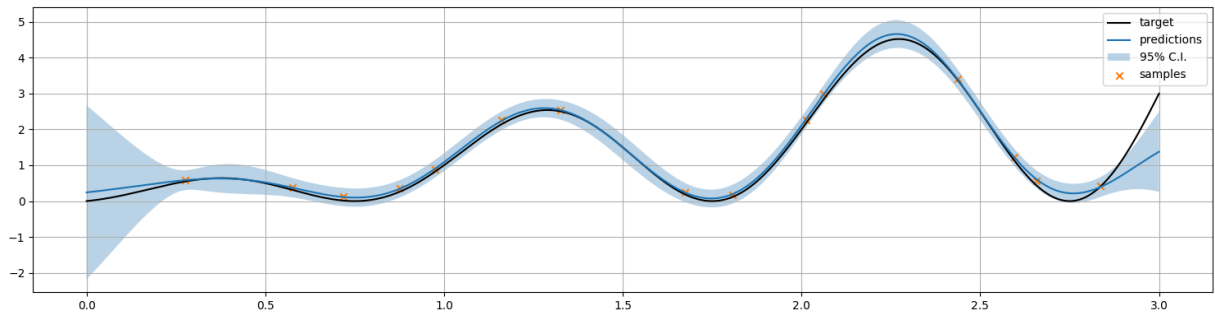
```
In [16]: gp = GaussianProcessRegressor(kernel=kernel, n_restarts_optimizer=9)
gp.fit(y_tr.index.values.reshape(-1,1), y_tr.values) # needs 2D input
print(gp.kernel_)
```

WhiteKernel(noise_level=0.01) + 2.21*2 * RBF(length_scale=0.321)

How to Improve the Model

Let us see the new predictions

```
In [17]: xp, std = gp.predict(x.reshape(-1,1), return_std=True)
util.plot_gp(target=y, samples=y_tr, pred=pd.Series(index=y.index, data=xp),
             std=pd.Series(index=y.index, data=std), figsize=figsize)
```



- Better, since the black curve is mostly in the confidence interval
- ...But we are still not exploiting the period and the trend

How to Improve the Model

So, let us take them into account, starting with the *period*

```
In [18]: from sklearn.gaussian_process.kernels import ExpSineSquared
kernel = WhiteKernel(0.1, (1e-2, 1e2))
kernel += ConstantKernel(1, (1e-2, 1e2)) * RBF(1, (1e-2, 1e2))
kernel += ExpSineSquared(1, 1, (1e-1, 1e1), (1e-1, 1e1))
```

ExpSineSquared captures the period:

$$K(x_i, x_j) = e^{-2 \frac{\sin^2\left(\pi \frac{d(x_i, x_j)}{p}\right)}{l^2}}$$

- The correlation grows if the distance is close to a multiple of the period p
- The scale parameter l controls the rate of decrease/increase
- In the implementation, the first parameter is l and the second p

How to Improve the Model

Now, let's try to capture the *trend*

```
In [19]: from sklearn.gaussian_process.kernels import DotProduct
kernel = WhiteKernel(0.1, (1e-2, 1e2))
```

```
kernel += ConstantKernel(1, (1e-2, 1e2)) * RBF(1, (1e-2, 1e2))
kernel += ExpSineSquared(1, 1, (1e-1, 1e1), (1e-1, 1e1))
kernel += DotProduct(1, (1e-2, 1e2))
```

DotProduct (somewhat) captures the trend:

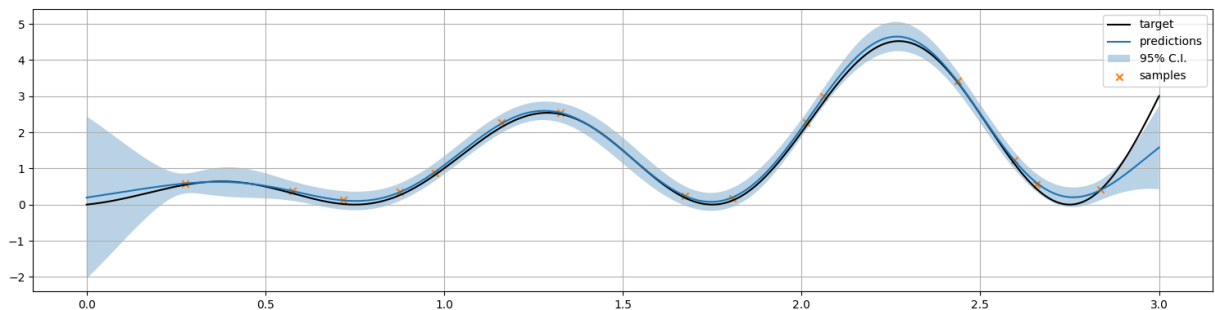
$$K(x_i, x_j) = \sigma^2 + x_i x_j$$

- The larger the x values, the larger the correlation
- This allows the distance from the mean (which is zero) to grow
- The σ parameter controls the base level of correlation
- Unlike all kernels so far **DotProduct** is *not translation-invariant*

How to Improve the Model

The new predictions are a bit better at the edges of the plot

```
In [20]: gp = GaussianProcessRegressor(kernel=kernel, n_restarts_optimizer=9)
gp.fit(y_tr.index.values.reshape(-1,1), y_tr.values) # needs 2D input
xp, std = gp.predict(x.reshape(-1,1), return_std=True)
util.plot_gp(target=y, samples=y_tr, pred=pd.Series(index=y.index, data=xp),
              std=pd.Series(index=y.index, data=std), figsize=figsize)
```



- Notice how the C.I. grows close to the sides?
- GPs tend to *interpolate well* and *extrapolate poorly*